

Section 5.3 Fundamental Theorem of Calculus

There are two main parts to the FTC.

I'll demonstrate each part with an example.

Ex ① Suppose $f'(x) = \frac{1}{1+36x^2}$. Find $f(x)$.

First rewrite $f'(x) = \frac{1}{1+(6x)^2}$

Make a guess: $\tan^{-1}$?

Try $f(x) = \tan^{-1}(6x) + C$.

Take the derivative to check the answer.

$$f'(x) = \frac{1}{1+(6x)^2} \cdot 6 \quad \leftarrow \text{chain Rule}$$

$$f'(x) = \frac{6}{1+36x^2}$$

The original $f'(x)$ did NOT have a 6.

So, adjust $f(x)$ by multiplying by $\frac{1}{6}$.

$$f(x) = \frac{1}{6} \cdot \tan^{-1}(6x) + c$$

Check it! $f'(x) = \frac{1}{6} \cdot \frac{1}{1+(6x)^2} \cdot 6$

$$f'(x) = \frac{1}{1+36x^2} \quad \checkmark$$

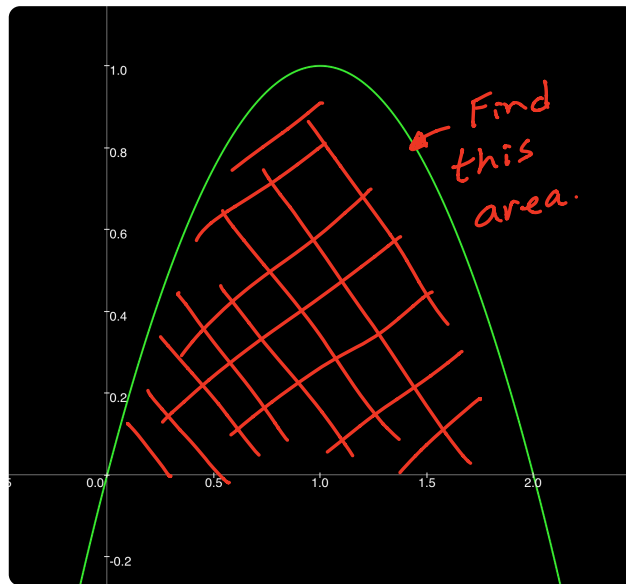
Notation $\int \frac{1}{1+36x^2} dx$ means:

Suppose $f'(x) = \frac{1}{1+36x^2}$. Find $f(x)$.

Conclusion $\frac{d}{dx} \left[\int \frac{1}{1+36x^2} dx \right] = \frac{1}{1+36x^2}$

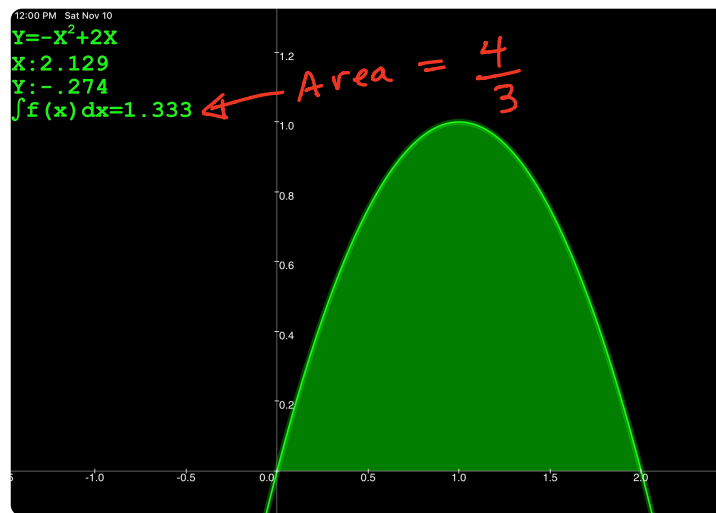
So, integration and differentiation are inverse operations.

Ex ② Find the area between $y = -x^2 + 2x$ and the x-axis on $[0, 2]$



$$\begin{aligned} \text{Area} &= \int_0^2 (-x^2 + 2x) dx \\ &= \left(-\frac{x^3}{3} + x^2 \right) \Big|_0^2 \\ &= \left(-\frac{2^3}{3} + 2^2 \right) - \left(-\frac{0^3}{3} + 0^2 \right) \\ &= -\frac{8}{3} + 4 = \left(\frac{4}{3} \right) \end{aligned}$$

Check it!



Conclusion To find Area under $f(x)$ on $[a, b]$,
use the antiderivative $F(x)$ and
find $F(b) - F(a)$.

Ex ③ Find $\frac{d}{dx} \left[\int_1^x \frac{1}{t} dt \right]$

Work inside the brackets, finding
the antiderivative of $\frac{1}{t}$ first.

$$= \frac{d}{dx} \left[\ln(t) \Big|_1^x \right]$$

$$= \frac{d}{dx} [\ln x - \ln 1]$$

$$= \frac{d}{dx} [\ln x - 0]$$

$$= \frac{1}{x}$$

④ Evaluate $\frac{d}{dx} \left[\int_1^{x^2} e^t dt \right]$

$$= \frac{d}{dx} \left[e^t \Big|_1^{x^2} \right]$$

$$= \frac{d}{dx} [e^{x^2} - e^1]$$

$$= e^{x^2} \cdot 2x \quad \leftarrow \text{chain Rule}$$

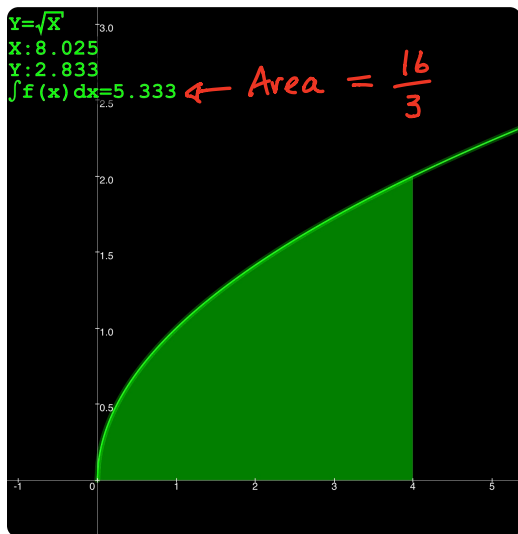
Ex ⑤ Find the area between $y = \sqrt{x}$ and the x-axis on $[0, 4]$.

$$\text{Area} = \int_0^4 \sqrt{x} \, dx = \int_0^4 x^{\frac{1}{2}} \, dx$$

$$= \left. \frac{2}{3} x^{\frac{3}{2}} \right|_0^4$$

$$= \frac{2}{3} (4)^{\frac{3}{2}} - \frac{2}{3} (0)^{\frac{3}{2}}$$

$$= \frac{2}{3} \cdot \sqrt{4}^3 = \frac{2}{3} \cdot 8 = \boxed{\frac{16}{3}}$$



Ex ⑥ $\int_1^2 (x+4)(x+1) \, dx$

We do NOT have the product rule for integrals (yet), so we need to start by multiplying this out.

$$(x+4)(x+1) = x^2 + 5x + 4$$

$$= \int_1^2 (x^2 + 5x + 4) dx$$

$$= \left(\frac{x^3}{3} + \frac{5x^2}{2} + 4x \right) \Big|_1^2$$

$$= \left(\frac{2^3}{3} + \frac{5(2)^2}{2} + 4(2) \right) - \left(\frac{1^3}{3} + \frac{5(1)^2}{2} + 4(1) \right)$$

$$= \left(\frac{8}{3} + \frac{20}{2} + 8 \right) - \left(\frac{1}{3} + \frac{5}{2} + 4 \right)$$

$$= 13.833$$

